In such heterogeneous equilibria, the concentration of the solid is a constant and is omitted in the above expression

$$K_{sp} = [X^{n+}]^m\ [Y^{m-}]^n$$

The solubility product of a compound is defined as the product of the molar concentration of the constituent ions, each raised to the power of its stoichiometric co – efficient in a balanced equilibrium equation.

Solubility product finds useful to decide whether an ionic compound gets precipitated when solution that contains the constituent ions are mixed.

When the product of molar concentration of the constituent ions i.e., ionic product, exceeds the solubility product then the compound gets precipitated.

The expression for the solubility product and the ionic product appears to be the same but in the solubility product expression, the molar concentration represents the equilibrium concentration and in ionic product, the initial concentration (or) concentration at a given time 't' is used.

In general we can summarise as,

Ionic product > K_{sp}, precipitation will occur and the solution is super saturated.

Ionic product < K_{sp}, no precipitation and the solution is unsaturated.

Ionic product = K_{sp}, equilibrium exist and the solution is saturated.

Example 8.9

Indicate find out whether lead chloride gets precipitated or not when 1 mL of 0.1M lead nitrate and 0.5 mL of 0.2 M NaCl solution are mixed? K_{sp} of $PbCl_2$ is 1.2×10^{-5}.

$$PbCl_2^{(s)} \underset{}{\overset{H_2O}{\rightleftharpoons}} Pb^{2+}(aq) + 2Cl^-(aq)$$

Ionic product = $[Pb^{2+}][Cl^-]^2$

Total volume = 1.5 mL

$$\underset{0.1M}{Pb(NO_3)_2} \rightleftharpoons \underset{0.1M}{Pb^{2+}} + 2NO_3^-$$

No of moles of Pb^{2+} = Molarity × volume of the solution in litre

$$= 0.1 \times 1 \times 10^{-3} = 10^{-4}$$

$$[Pb^{2+}] = \frac{\text{number of moles of } Pb^{2+}}{\text{Volume of the solution in L}} = \frac{10^{-4}}{1.5 \times 10^{-3}\ mL} = 6.7 \times 10^{-2} M$$

$$\underset{0.2M}{NaCl} \longrightarrow \underset{0.2M}{Na^+} + \underset{0.2M}{Cl^-}$$

No of moles of Cl^- = $0.2 \times 0.5 \times 10^{-3} = 10^{-4}$

$$[Cl^-] = \frac{10^{-4}\ \text{moles}}{1.5 \times 10^{-3}\ L} = 6.7 \times 10^{-2} M$$

Ionic product = $(6.7 \times 10^{-2})(6.7 \times 10^{-2})^2 = 3.01 \times 10^{-4}$

Since, the ionic product 3.01×10^{-4} is greater than the solubility product (1.2×10^{-5}), $PbCl_2$ will get precipitated.

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